

AMARNATH S. PATEL

Orlando, FL | amarnath.patel@ucf.edu | apatel.co | github.com/jeebuscrossaint | [ORCID 0009-0008-9460-082X](https://orcid.org/0009-0008-9460-082X)

EDUCATION

University of Central Florida **Undergraduate Student 4.00 GPA**
B.S. in Physics (Optics & Lasers) and Mathematics, Computer Science Minor (38 credit hours) August 2025 – Present

Florida Atlantic University **3.66 GPA**
University coursework completed via FAU High School, ages 14–18 (111 credit hours) August 2021 – May 2025

RESEARCH EXPERIENCE

UCF Astrophotonics Lab, CREOL – Undergraduate Researcher August 2025 – Present
Supervisor: Dr. Stephen Eikenberry

- Conduct the measurement and analysis of the wavelength-dependent complex transfer matrix of a photonic lantern by [off-axis digital holography](#): perform the optical alignment and bench acquisition, and built the automated all-port $\times$ C-band (1525–1575 nm) sweep that runs unattended (extending Dobias et al., *Opt. Express* **34**(9), 17217 (2026)).
- Developed the full phase-retrieval chain – FFT sideband isolation and demodulation, Butterworth low-pass, joint optimization of mode-field diameter, quadratic-phase correction and position, then LP-mode decomposition recovering complex modal amplitude and phase at $\approx 98\%$ fidelity – and produced the per-port fidelity characterization for 6- and 7-port lanterns.
- Integrated and control the four-instrument measurement chain across three protocols: HP 8168E tunable laser (GPIB/VISA), Xenics Bobcat 320 InGaAs camera (GigE Vision), DiCon GP700 fiber switch (RS-232), and Thorlabs MPC320 polarization controller (USB/Kinesis), with automated polarization optimization for peak fringe contrast and saturation rejection.
- Developed a Python SDK wrapper ([gentec-camera](#)) for an IR beam-profiling camera enabling automated acquisition, real-time beam analysis, and FITS output.
- Contribute to [PolyOculus](#), an astronomical instrument program building photometric observing capability from a networked array of 8 telescopes: developed INDI-protocol mount control, focuser automation, RA/Dec slewing, and backlash compensation.

UCF Physics Department – Undergraduate Research Assistant March 2026 – Present
Physics Education Research – Dr. Zhongzhou Chen (NSF Award 2421299; Gates Foundation INV-076932)

- Research on automated assessment generation for introductory physics instruction: built [ESTELA](#), a system producing multi-version isomorphic exams from structured problem banks, and an agentic generation–evaluation pipeline that iteratively drafts and validates new problems against instructor criteria.
- Authored the documentation and distribution materials used at a faculty workshop, and maintain the public release pipeline that external adopters depend on.
- Lead developer across two independent implementations of the assessment tool, including technical ownership of a parallel version authored by another group member. Manuscript in preparation.

FAU Grant-Funded AI Safety Research Project (High School) January 2024 – March 2025
Supervisor: Tucker Hindle, Florida Atlantic University

- Benchmarked 10+ coherence and detection methods (GPT-2 perplexity, RoBERTa, BERT NSP, LSA, NLI, burstiness) across 3,125+ generated sequences; identified GPT-2 perplexity as the strongest discriminator ($3.3\times$ separation between coherent and incoherent text).
 - Characterized a consistent quality–evasion tradeoff across four adversarial humanization pipelines. Grant-funded with HPC access; presented at the Wilkes Honors College Symposium (2025).
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OTHER EXPERIENCE

Secretary – National Society of Black Physicists (NSBP), UCF Chapter 2025 – Present

- Coordinate meetings, records, and communications for the UCF student chapter of the National Society of Black Physicists.

President — Coding Club, Florida Atlantic University (High School) 2023 – 2025

- Founded and led a student coding club; organized meetings and collaborative programming projects with peers.

Teaching Assistant — Calculus (High School) August 2024 – May 2025

- Assisted 70 undergraduate students with learning calculus; office hours, exam review, grading. Part-time (10 h/week).

Advanced Experimental Vehicles — Programmer, Leader, Builder (High School) November 2023 – May 2025

- Configured Arch Linux ARM on Raspberry Pi 5 with Hyprland compositor and WireGuard VPN, enabling worldwide real-time telemetry monitoring of BMS, GPS, and camera feeds.
- Won 2nd Place in Division and Lockheed Martin Award for “Highest Level of Engineering Excellence.”

PRESENTATIONS

Complex Transfer-Matrix Characterization of a Photonic Lantern by Off-Axis Digital Holography November 2026 (upcoming)

Joint Annual Conference of the National Society of Black Physicists and the National Society of Hispanic Physicists, North Carolina.

Batch Processing for Automated Grading via Azure OpenAI June 2026

University of Central Florida, Downtown Campus. ESTELA project (Physics Education Research, Dr. Zhongzhou Chen).

Coherence and Detection Approaches for Identifying AI-Generated Text March 2025

Wilkes Honors College Undergraduate Research Symposium, Florida Atlantic University.

TECHNICAL REPORTS

Coherence and Detection Approaches for Identifying AI-Generated Text 2025

- Empirical study of AI-text detection and adversarial evasion: benchmarked 10 coherence and detection methods (GPT-2 perplexity best, 3.35× separation; RoBERTa the only effective detector) and built four evasion pipelines, characterizing a consistent quality–evasion tradeoff. Grant-funded, Florida Atlantic University.

SELECTED PROJECTS

CELERIS — Metalens Design Pipeline & From-Scratch RCWA Solver 2026

- From-scratch solver for Maxwell’s equations in periodic subwavelength structures via rigorous coupled-wave analysis / the Fourier modal method: 1D TE/TM multilayer gratings with Li/Liu–Fan inverse-rule factorization, full 2D-vectorial $P \cdot Q$ formulation for bi-periodic nanopillar arrays, and stable Redheffer scattering-matrix layer recursion.
- Validated every layer of the stack against closed-form physics, an independent method, or energy conservation: RCWA vs. transfer-matrix method to 1e-6, multilayer S-matrix to 1e-15, 2D→1D degenerate reduction to 5e-12, Σ (diffraction efficiency) = 1.000000, and diffraction-limited designed focal spots at $\text{FWHM} = \lambda f / D$. Independently cross-validated against the grcwa and Stanford S⁴ solvers ($\approx 1e-7$) and reproduces canonical published metalenses (Khorasaninejad et al., *Science* 2016; Chen et al., *Nat. Nanotechnol.* 2018).
- Implemented the physical design and analysis capability standalone RCWA kernels lack: Sellmeier and tabulated n, k dispersion, meta-atom library sweeps with automatic etch-depth selection for full- 2π phase coverage, gradient-based (Adam) inverse design, Pancharatnam–Berry geometric-phase and polarization-multiplexed lenses exploiting form birefringence (reporting per-polarization RMS phase and focal isolation in dB), plus Strehl, Zernike wavefront, MTF, chromatic focal shift, through-focus caustic, and fabrication-tolerance Monte-Carlo analysis with fabrication-ready GDSII export.

- Accelerated Rayleigh–Sommerfeld far-field propagation as a CUDA kernel: 675× over a 16-core CPU at 92k pillars (23.9 s → 35 ms), agreeing to $\approx 1e-5$, with automatic CPU fallback. C++23/CUDA, with Python bindings and a native desktop GUI. *Sole-author manuscript in preparation (Computer Physics Communications).*

TECHNICAL SKILLS

Programming Languages: C/C++, CUDA, Rust, Python, JavaScript, Shell (Fish, Bash, tcsh), LaTeX/Typst

Scientific Computing: NumPy, SciPy, GPU/CUDA acceleration, FFT & signal processing, electromagnetic simulation (RCWA)

Tools & Libraries: Qt/PySide6, Docker, Git, CMake, XMake, INDI, KaTeX

Instrumentation: GPIB/VISA, RS-232, GigE Vision camera acquisition, tunable IR laser control, motorized polarization control, FITS & GDSII data formats

Operating Systems: Linux (Arch, NixOS, Ubuntu), BSD, Windows

RELEVANT COURSEWORK

University of Central Florida

Physics

Geometric Optics & Lab
Quantum Information Processing
Independent Research [PHY 4912] (In Progress)

Modern Physics [PHY 3101]
Mathematical Methods for Physics [PHZ 3113]
Electricity & Magnetism I (In Progress)

Mathematics

Applied Linear Algebra [MAS 3105]
Linear Algebra (Proof-Based) [MAS 3106] (In Progress)

Complex Analysis [MAA 4402]
Partial Differential Equations (In Progress)

Computer Science

C Programming [EGN 3211]
Object-Oriented Programming (In Progress)

Discrete Structures

Florida Atlantic University

Physics

General Physics I

General Physics II — Honors

Mathematics

Calculus I
Honors Calculus III
Elementary Matrix Algebra [MAS 2103]

Calculus II
Ordinary Differential Equations [MAP 2302]

Computer Science

Data Structures & Algorithms
Computer Architecture
C++ Programming [COP 3014]

Computer Logic Design
Deep Learning [CAP 4613]
Web Development [COP 3813]

HONORS & AWARDS

1st Place, Night Hacks Hackathon — UniUtils (course schedule generator & classroom finder)	2023
Lockheed Martin Award — “Highest Level of Engineering Excellence,” AEV Competition	2024
2nd Place in Division, Advanced Experimental Vehicles Competition	2024
Florida Bright Futures — Florida Academic Scholars (highest tier; 100% tuition)	2025
Barry M. Goldwater Scholarship — UCF institutional nominee	2026
Astronaut Scholarship Foundation — UCF institutional nominee	2026